

The role of m6A RNA methylation in the maintenance of X chromosome inactivation and X-to-autosome dosage compensation in early embryonic lineages

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SUMMARY

In mammals, while X chromosome inactivation (XCI) balances the dosage of X-linked gene expression between sexes, upregulation of active-X balances the dosage of monoallelic X-linked genes with biallelic autosomal genes (AA). Here, we have investigated the role of m6A RNA methylation in the maintenance of XCI and X-to-autosome (X-to-A) dosage compensation in early embryonic lineages: epiblast stem cells (EpiSCs), trophoblast stem cells (TSCs), and extraembryonic endoderm stem cells (XENs). We find that the depletion of m6A RNA methylation in these cells does not affect the maintenance of inactive-X silencing. Moreover, we show that m6A marks are less enriched on X-linked transcripts than the autosomal transcripts in early embryonic lineages. Notably, we demonstrate that the extent of X-to-A dosage compensation varies with m6A methylation level. Finally, we show that the depletion of m6A partly disrupts X-to-A dosage compensation in a cell-type-specific manner. Together, our study provides significant insight into the role of m6A RNA methylation in dosage compensation.

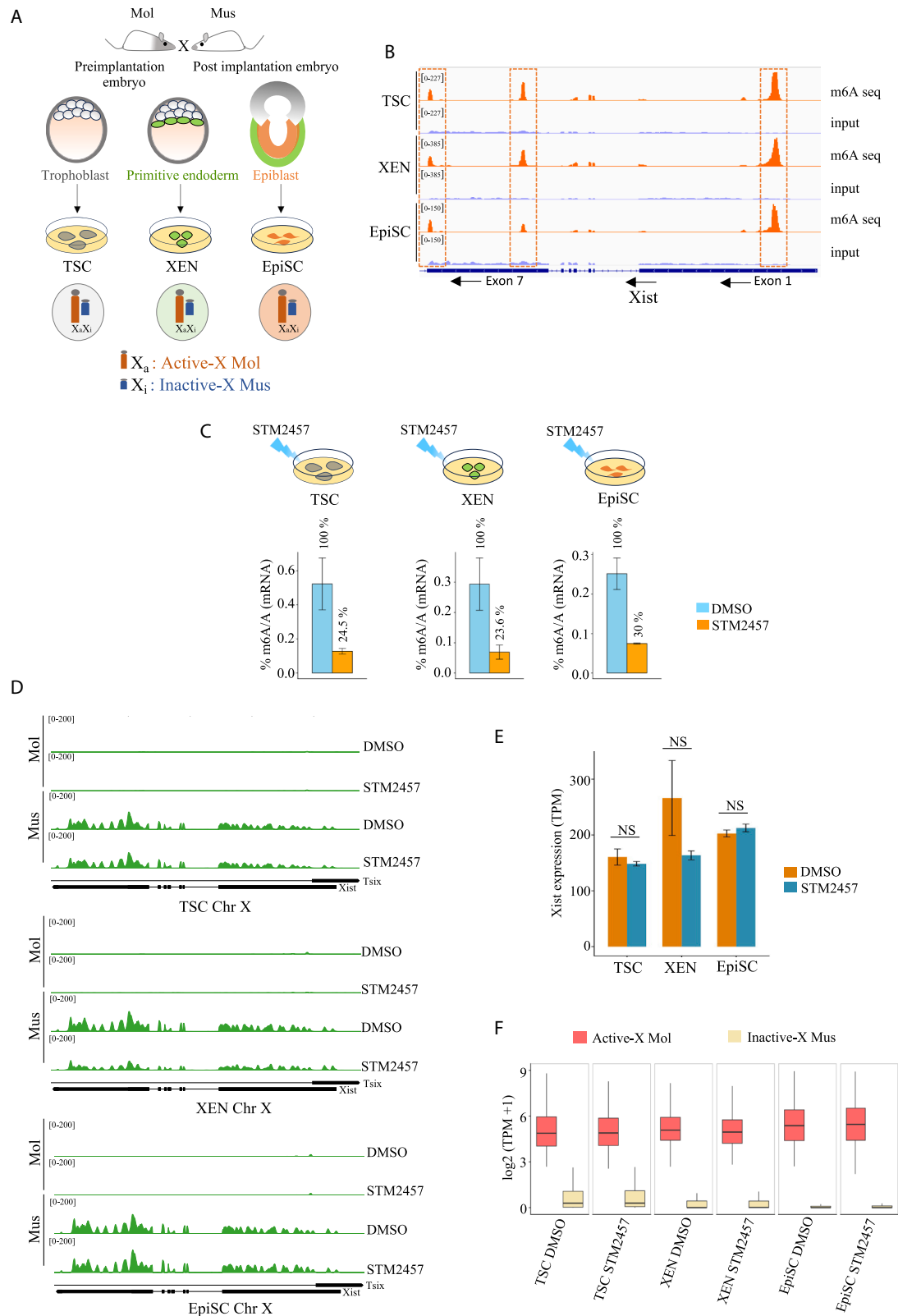
INTRODUCTION

In the course of evolution, the Y chromosome underwent massive degradation through gene-loss events (Bachtrog, 2013; Graves, 2016; Lahn and Page, 1999). This rendered male-specific monoallelic expression of X-linked genes in contrast to biallelic expression from the autosomes. Into this, Susumu Ohno hypothesized that the X chromosome in males underwent upregulation to balance the X-to-autosome (X-to-A) dosage (Ohno, 1967). Subsequently, the inheritance of this upregulated-X in females is thought to have created an overdose of X-linked gene products, which was rectified by the evolution of X chromosome inactivation (XCI) mechanisms (Lyon, 1961). In recent years, the mechanisms of XCI have been investigated extensively. However, the mere existence of X chromosome upregulation (XCU) in mammals has been contested for decades. Some studies refuted Ohno's hypothesis (Chen et al., 2020; Lin et al., 2012; Wang et al., 2017; Xiong et al., 2010; Yang and Chen, 2019). On the other hand, many studies provided evidence in support of XCU (Borensztein et al., 2017a; Cidral et al., 2021; Deng et al., 2011; 2013; Di and Disteche, 2006; Gupta et al., 2006; Kharchenko et al., 2011; Larsson et al., 2019; Lin et al., 2007; 2011; Mahadevaiah et al., 2020; Mandal et al., 2020; De Mello et al., 2017; Sangrithi et al., 2017; Wang et al., 2016; Yildirim et al., 2012). Importantly, recent

studies leveraging allele-resolved transcriptomic analysis at the single-cell level have clearly shown coordinated upregulation of the active-X chromosome upon silencing of genes on the inactive-X during early embryogenesis (Lentini et al., 2022; Naik et al., 2022; 2024). Together, while XCI balances the dosage of X-linked gene expression between sexes, XCU contributes to balancing the X-to-A dosage. However, the mechanisms and factors involved in maintaining XCI and X-to-A dosage compensation remain poorly understood.

N6-methyladenosine (m6A) is one of the most prevalent internal reversible modifications known to be present on RNAs (Dominissini et al., 2012; Meyer et al., 2012). Recently, m6A RNA methylation has emerged as a new player in gene regulation, which plays key roles in embryonic development, cellular differentiation, maintenance of cellular integrity, and stress response (Batista et al., 2014; Geula et al., 2015; H. Lee et al., 2019; Li et al., 2017a). Dysregulation of m6A has been attributed to many pathophysiological conditions, including cancer (Berulava et al., 2020; Jiang et al., 2021; Pupak et al., 2022). m6A modification is deposited on the RNAs by the core methyltransferase complex consisting of METTL3, which is the main catalytic methylase (writer); METTL14; and WTAP, along with accessory proteins (Liu et al., 2013; Ping et al., 2014). On the other hand, FTO and ALKBH5 act as an eraser for m6A RNA modification (Jia et al., 2011; Zheng et al.,





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2013). Finally, the “reader” proteins recognize the m6A mark on the RNAs and decide their fate within a cell (Knuckles et al., 2017; Shi et al., 2017). Several studies have shown that m6A deposition occurs mostly at the 3′ regions of transcripts and is believed to regulate translation efficiency, alternative splicing events, nuclear mRNA export, and mRNA turnover (Ke et al., 2015; Roundtree et al., 2017; Slobodin et al., 2017; Wang et al., 2013; 2015; Xiao et al., 2016). Although m6A-mediated regulation is known to majorly control RNA metabolism, it has also been shown to act as a global epigenetic regulator (Li et al., 2020; Selmi and Lanzuolo, 2022; Wei et al., 2022). Together, m6A RNA methylation has emerged as a key player in the epigenetic regulation of gene expression. Emerging studies suggest that m6A may play a critical role in regulating dosage compensation as well. However, the precise role of m6A RNA methylation in dosage compensation remains underexplored. Into that, X-inactive specific transcript (*Xist*) long non-coding RNA (lncRNA), which is the master regulator of XCI, has been shown to be highly enriched with m6A modification. Although some studies have implicated the role of m6A regulation in *Xist*-mediated gene silencing (Chang et al., 2022; Nesterova et al., 2019; Patil et al., 2016), the essentiality of m6A in maintaining XCI remains unknown. On the other hand, upregulation of the active-X has been reported at the transcriptional level through enrichment of active chromatin marks, increased chromatin accessibility, and increased transcriptional burst kinetics of X-linked transcripts (Deng et al., 2013; Larsson et al., 2019; Talon et al., 2021). Given the versatile nature of m6A acting as a global epitranscriptomic regulator, it is thought that m6A-mediated regulation of transcript stabilization contributes to equilibrate the X-to-A dosage. Indeed, a recent study has shown that m6A RNA methylation contributes partly to the X-to-A dosage compensation (Rücklé et al., 2023). However, the potential contribution of m6A in fine-tuning X-to-A dosage across different cell types and developmental contexts remains underexplored. In this study, we have explored the relative contribution of m6A in maintaining XCI as well as in maintaining an equilib-

rium X-to-A dosage in different early embryonic lineages and other cell types.

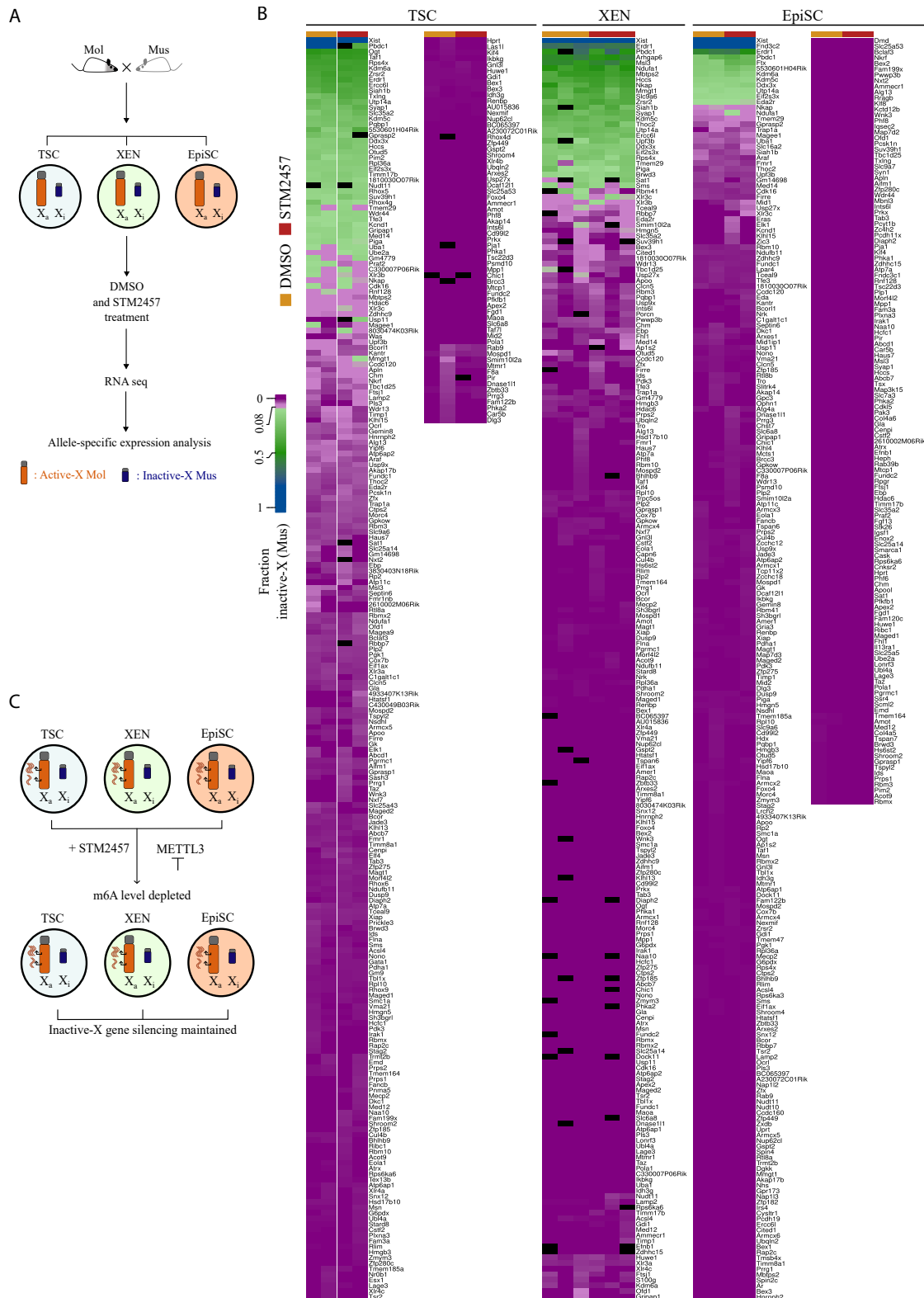
RESULTS

Depletion of m6A RNA methylation does not perturb the maintenance of X-linked gene silencing in XENs, TSCs, and EpiSCs

Mouse undergoes two waves of XCI: imprinted inactivation (iXCI) of the paternal-X at the 2- to 4-cell stage of embryogenesis followed by subsequent reactivation of the paternal-X in the inner cell mass of the late blastocyst and finally, random inactivation (rXCI) of either of the paternal- or maternal-X in the embryonic epiblast (Du et al., 2024; Harris et al., 2019; Huynh and Lee, 2003; Maclary et al., 2014; Okamoto et al., 2004; Takagi and Sasaki, 1975). To investigate the role of m6A RNA methylation in the maintenance of XCI, we used stem cell lines representing early embryonic lineages. We have used extraembryonic endoderm cells (XENs) that represent the primitive endoderm of late blastocysts, which have already initiated their iXCI process and, thereby, maintain the iXCI state (Figure 1A). On the other hand, we have used trophoblast stem cells (TSCs) that represent the trophoctodermal cells of the blastocyst and are in a maintenance phase of iXCI (Figure 1A). Apart from these two, we have used epiblast stem cell (EpiSC) lines resembling much of post-implantation epiblast, which represents the maintenance phase of rXCI (Figure 1A). Notably, all are hybrid cell lines carrying polymorphic X chromosomes (X^{Mus} : *Mus musculus* origin and X^{Mol} : *Mus molossinus* origin) and in all cell lines, the X^{Mus} is selectively inactivated, which allowed us to disentangle the X-linked gene expression between active- vs. inactive-X through allele-specific analysis (Figure 1A). First, we delineated if m6A RNA modification is present in the *Xist* RNA expressing in XENs, EpiSCs, and TSCs by performing methylated RNA immunoprecipitation sequencing (MeRIP-seq). We find that m6A modifications are present at exons 1 and 7 of *Xist* RNA in TSCs, XENs, and EpiSCs (Figure 1B). Previously, it was implicated

Figure 1. Profiling the effect of m6A depletion on *Xist* expression and maintenance of X-linked gene silencing in female TSCs, XENs, and EpiSCs

- (A) Schematic showing the hybrid TSCs, XENs, and EpiSC lines carrying polymorphic X chromosomes (X^{Mus} : *Mus musculus* origin and X^{Mol} : *Mus molossinus* origin). X^{Mus} is inactivated in these cells.
 (B) IGV browser snapshot of m6A sites in *Xist* in TSCs, XENs, and EpiSCs; dotted lines represent m6A sites.
 (C) Quantification of m6A level relative to A in mRNA of TSCs, XENs, and EpiSCs by LC-MS/MS upon METTL3 inhibitor (STM2457, 50 μ M) and vehicle (DMSO) treatment for 12 h. Error bar represents standard deviation (SD), TSCs ($n = 2$), XENs ($n = 3$), and EpiSCs ($n = 2$).
 (D) Tracks showing the allelic *Xist* expression in DMSO- and STM2457-treated TSCs, XENs, and EpiSCs.
 (E) Quantification of *Xist* expression in DMSO- vs. STM2457-treated TSCs, XENs, and EpiSCs; two-sample t test; NS, non-significant.
 (F) Boxplot showing the expression of X-linked genes from active-X (X^{Mol}) and inactive-X (X^{Mus}) in DMSO- vs. STM2457-treated TSCs, XENs, and EpiSCs.



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that m6A RNA modification in *Xist* RNA contributes to *Xist*-mediated gene silencing (Chang et al., 2022; Patil et al., 2016). Moreover, m6A RNA modification can contribute to maintaining *Xist* RNA expression level (Wei et al., 2025). Therefore, we investigated whether m6A depletion affects *Xist* expression and *Xist*-mediated gene silencing. Of note, it has been demonstrated that *Xist* is largely dispensable for maintaining X-linked gene silencing; only a subset of X-linked genes has been found to depend on *Xist* to maintain their silencing status (Arava et al., 2023; Richart et al., 2022; Yang et al., 2016, 2022). Therefore, we predicted that m6A depletion in XENs, TSCs, and EpiSCs may have a minor effect on *Xist*-mediated X-linked gene silencing. To explore this, we depleted m6A in XENs, TSCs, and EpiSCs through inhibition of m6A methyl transferase METTL3 using STM2457 inhibitor (Yankova et al., 2021) (Figure 1C). To validate the m6A depletion upon inhibition of METTL3, we performed liquid chromatography-tandem mass spectrometry (LC-MS/MS) to quantify m6A levels in mRNAs isolated from DMSO-treated and inhibitor (STM2457)-treated XENs, TSCs, and EpiSCs. We observed robust m6A depletion upon inhibition of METTL3 (XENs, $n = 3$; EpiSCs and TSCs, $n = 2$) (Figure 1C). Notably, in the inhibitor-treated cells, we observed a 76.4% reduction of m6A/A levels in XEN, 75.5% in TSCs, and 70% in EpiSCs compared to DMSO treatment (Figure 1C). Next, we performed RNA sequencing (RNA-seq) of DMSO- vs. STM2457-treated XENs, TSCs, and EpiSCs. From the RNA-seq data, we compared the *Xist* expression level between DMSO- vs. inhibitor-treated cells. We find that *Xist* RNA is exclusively expressed from the X^{Mus} allele in all three cell types, thereby validating that X^{Mus} is the inactive-X (Figure 1D). Notably, *Xist* expression remains unaltered between DMSO- vs. inhibitor-treated TSCs and EpiSCs (Figures 1D and 1E). However, we observed that *Xist* expression was slightly reduced in XENs upon inhibitor treatment (Figures 1D and 1E). Together, we conclude that m6A depletion does not affect the *Xist* RNA level in TSCs and EpiSCs; however, it leads to a reduction in *Xist* expression in XENs. Next, we explored if m6A depletion affects the X-linked gene silencing. To test this, we performed allele-specific expression analysis of the RNA-seq data. We observed exclusive expression of X-linked genes from the active-X (X^{Mol}) in all XENs, TSCs, and EpiSCs of DMSO-treated groups, further validating that X^{Mus} is the

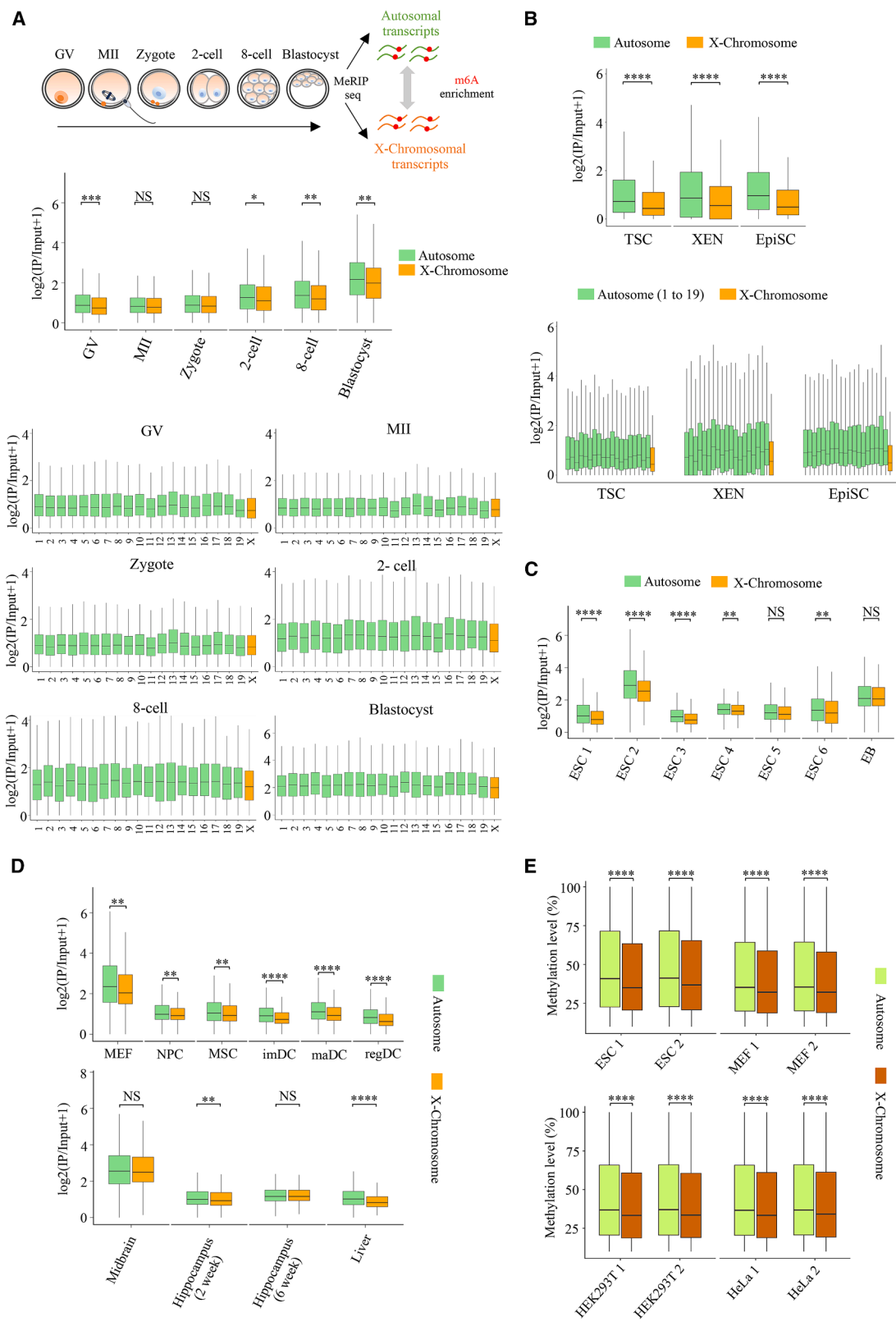
inactive-X in these cell lines (Figure 1F). Upon inhibitor treatment, there were no overall changes in inactive-X expression compared to the DMSO-treated cells, indicating no effect on X-linked gene silencing despite the m6A depletion (Figure 1F). To probe this further, we performed a chromosome-wide expression analysis of all the X-linked genes in XENs, TSCs, and EpiSCs at an allelic resolution (Figure 2A). As expected, we observed majority of the X-linked genes from the inactive-X (X^{Mus}) remained silenced in the DMSO-treated XENs, TSCs, and EpiSCs (Figure 2B). Only a few genes showed biallelic expression, which are known escapees of XCI (Berletch et al., 2011) (Figure 2B). Surprisingly, we observed absolutely no signs of reactivation of inactivated genes upon m6A depletion in the inhibitor-treated XENs, TSCs, and EpiSCs (Figure 2B). Only in TSCs we observed reactivation of a few genes (*Ube2a*, *Nkap*, *Xlr3C*, *Usp11*, *Mmgt1*, *8030474K03Rik*). Among these genes *Ube2a* gene has been reported to be reactivated upon ablation of *Xist* in blood cells (B cells) (Yildirim et al., 2013). Additionally, we also observed reactivation of the gene *Ndufa1* in EpiSCs (Figure 2B). In contrast, we also observed inactivation of previously active genes in TSCs (*Praf2*) and XENs (*Rbm41*). Overall, our analysis suggests that m6A RNA methylation is dispensable for maintaining the X-linked gene silencing in female XENs, TSCs, and EpiSCs (Figure 2C). It is worth to mention that 12-h inhibition may not be sufficient to capture the genes that are prone to be reactivated late (Borensztein et al., 2017b; Janiszewski et al., 2019; Naik et al., 2024; Roidor et al., 2025; Talon et al., 2021); therefore, in future, prolonged depletion of m6A may provide better clarity on the requirement of m6A in maintaining X-linked gene silencing.

m6A is less enriched in X-linked transcripts compared to the autosomal transcripts in early embryonic lineages

Differential m6A enrichments in the transcripts are known to modulate the transcript stability and balance the gene expression stoichiometry (Zaccara and Jaffrey, 2020). One of the prevailing hypotheses is that the higher stability of X chromosome transcripts relative to autosomal transcripts partially contributes to the balance of the X-to-A expression dosage in cells. Recently, it has been demonstrated that the enrichment of m6A in X-linked transcripts is lower than that in autosomal transcripts, resulting in better

Figure 2. Depletion of RNA m6A level does not affect the maintenance of XCI

(A) Schematic workflow showing RNA-seq analysis in the DMSO- and STM2457-treated TSCs, XENs, and EpiSCs at allelic resolution. (B) Heatmap showing the fraction of expression of X-linked genes from inactive-X (X^{Mus}) in DMSO- vs. STM2457-treated TSCs, XENs, and EpiSCs. Heatmaps for TSCs and EpiSCs are divided into two parts, but they represent expression from a single X chromosome. (C) Model diagram showing that despite m6A depletion, X-linked gene silencing on the inactive-X is maintained in STM2457-treated TSCs, XENs, and EpiSCs.



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stability of X-linked transcripts (Rücklé et al., 2023). However, the pattern of m6A enrichment in X-linked and autosomal transcripts during early embryogenesis remains unknown. To explore this, we profiled the m6A enrichment on X-linked vs. autosomal transcripts during early embryogenesis using available MeRIP-seq datasets (Wang et al., 2023b) (Figure 3A). Interestingly, we find that m6A modifications are less enriched in X-linked transcripts compared to the autosomal transcripts in germinal vesicle-stage (GV) oocyte (Figure 3A). However, we did not find any significant differences in X-linked vs. autosomal m6A enrichments in metaphase II (MII) oocyte and zygote (Figure 3A). But, we observed that enrichment of m6A in X-linked transcripts is reduced compared to the autosomal transcripts starting from 2-cell stage to 8-cell and blastocyst stage (Figure 3A). Furthermore, to gain better clarity on m6A enrichment in the developmental milieu, we performed m6A enrichment analysis using common transcripts present across all developmental stages and observed a similar pattern to that above (Figure S1A). Next, we extended our analysis to different embryonic lineages. For this, we performed MeRIP-seq in TSCs, XENs, and EpiSCs lines and found that m6A modification is significantly less enriched in X-linked transcripts compared to the autosomal transcripts in these cells as well (Figure 3B). Similar pattern was observed in different mouse embryonic stem cell (ESC) lines, but not much differences was observed in embryoid bodies, a three-dimensional aggregation formed by ESCs *in vitro* (Figure 3C). Furthermore, we noticed similar trend in other cell types such as mouse embryonic fibroblasts (MEFs), neural progenitor cells (NPCs), immature dendritic cells, mature dendritic cells (maDCs), regulatory dendritic cells, and mesenchymal stem cells (MSCs) (Figure 3D). In contrast, of the three tissue types that we analyzed, difference in m6A levels was not observed in the midbrain and hippocampus (6 weeks); however, m6A was lower on X-linked transcript compared to the autosomes in the hippocampus (2 weeks) and liver tissue

(Figure 3D). Furthermore, we extended our analysis using available GLORI (glyoxal and nitrite-mediated deamination of unmethylated adenosines) data, which provides quantitative estimation of m6A enrichments (Liu et al., 2023). We found that X-linked transcripts have a lower methylation level compared to autosomal transcripts in ESCs, MEFs and HEK293T and HeLa cells (Figure 3E). Taken together, we conclude that m6A is less enriched in X-linked transcripts compared to the autosomal transcripts in early embryonic lineages as well as in somatic lineages, with exceptions to certain somatic tissues.

The extent of X-to-autosome dosage compensation varies with the m6A methylation level of X-linked transcripts

Next, we were curious to know if m6A RNA methylation contributes to the maintenance of X-to-A dosage compensation, as m6A was differentially enriched between X-linked and autosomal transcripts (Figure 3). It is believed that less enrichment of m6A on X-linked transcripts increases their stability and thereby partly contributes to the maintenance of the X-to-A dosage equilibrium. To explore this further, we compared the X-to-A dosage compensation pattern between m6A methylated and non-methylated X-linked transcripts. To profile the X-to-A dosage compensation, we calculated the allelic X-to-autosome (X:A) expression ratio using a bootstrap method, where an equal number of autosomal genes with respect to X-linked genes were randomly sampled 1,000 times (Borensztein et al., 2017a; Lentini et al., 2022; Naik et al., 2022; 2024; Pacini et al., 2021). We excluded genes with low expression (<5 TPM [transcripts per million]) and those with very high expression (above the 95th percentile) for the X:A analysis. If there is X-to-A dosage compensation, the active- $X^{Mol}:A^{Mol}$ ratio is expected to be >1 and close to 2. Indeed, we find that active- $X^{Mol}:A^{Mol}$ ratio is ~2 in XENs and TSCs; however, for EpiSCs, we observed the ratio is slightly higher than 2 (Figure 4A). Estimation of X:A ratio using a common set of genes across

Figure 3. Comparative profiling of m6A enrichment between autosomal and X chromosome transcripts at different stages of early embryogenesis and different embryonic lineages

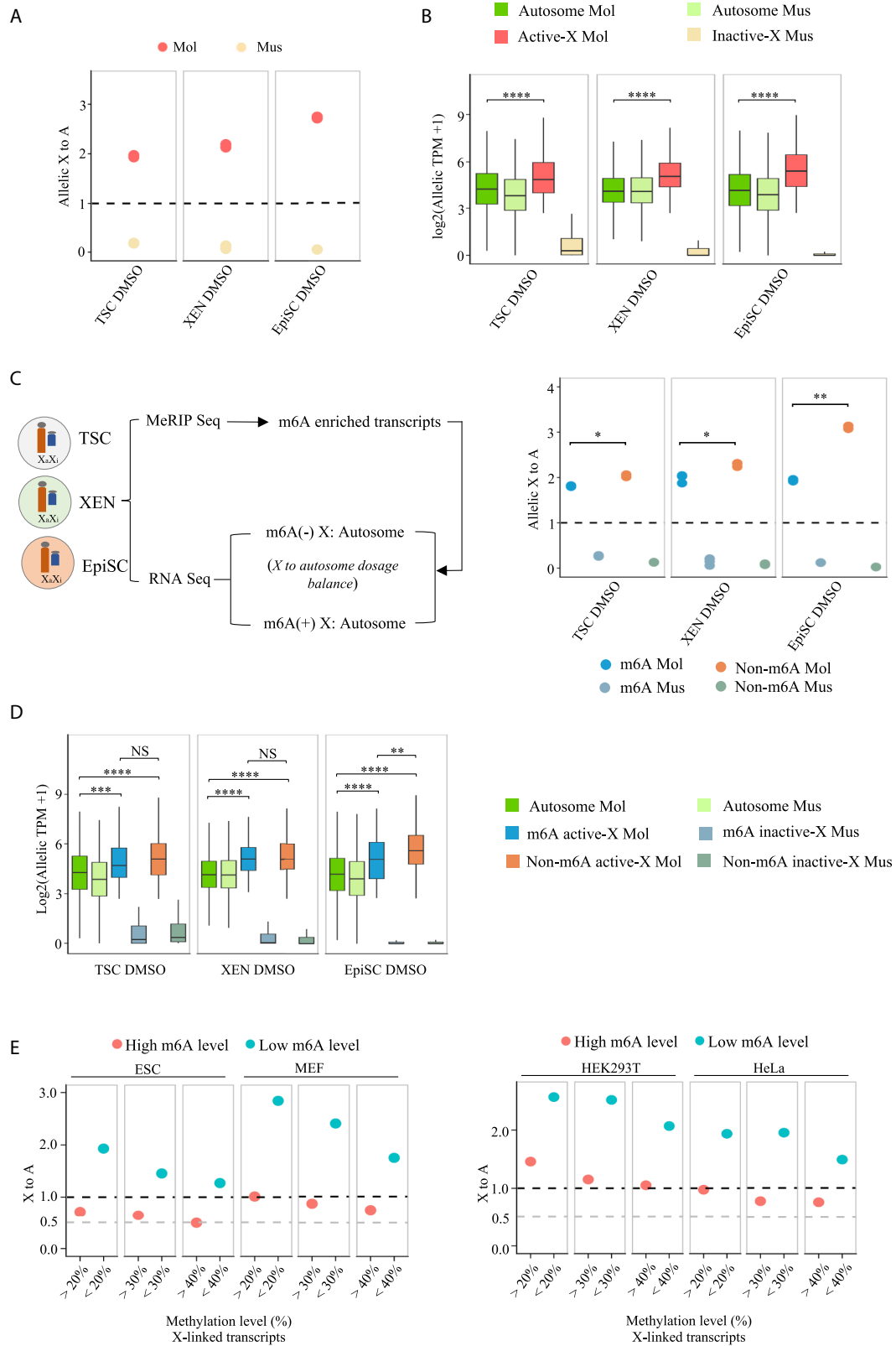
(A) Schematic showing the workflow of m6A enrichment analysis in autosomal and X-linked transcripts during early embryogenesis (top); RNA m6A enrichment autosomes vs. X-linked transcripts in mouse oocyte and various stages of mouse early embryonic development (middle). Chromosome-wise RNA m6A enrichment (bottom). Edges of each box represent the 25th and 75th quartiles, and the center line denotes the median value; Mann-Whitney U test; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; NS, nonsignificant.

(B) RNA m6A enrichment in autosomal vs. X chromosome transcripts in TSCs, XENs, and EpiSCs (top). Mann-Whitney U test, **** $p < 0.0001$; chromosome-wise RNA m6A enrichment in TSCs, XENs, and EpiSCs (bottom).

(C) Autosome vs. X chromosome transcripts' RNA m6A enrichment in mouse ESCs and EB (embryoid body); Mann-Whitney U test, ** $p < 0.01$, **** $p < 0.0001$; NS, nonsignificant.

(D) Autosome vs. X chromosome transcripts' m6A enrichment in different somatic cell types (top), and tissues (bottom). Mann-Whitney U test, ** $p < 0.01$, **** p value <0.0001; NS, nonsignificant.

(E) Quantification of m6A methylation between autosomal vs. X chromosome transcripts using GLORI method among different cell lines: (top) mouse (ESCs and MEFs) and (bottom) human (HEK293T and HeLa). Mann-Whitney U test, **** $p < 0.0001$.



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XENs, TSCs, and EpiSCs also showed a similar trend (Figure S1B). We conclude that XENs, TSCs, and EpiSCs undergo robust X-to-A dosage compensation. To test further, we compared the overall allelic expression of autosomal and X-linked genes. We observed that the expression of X-linked genes from the active-X was significantly higher compared to the autosomal allelic expression in XENs, TSCs, and EpiSCs, corroborating the upregulation of X-linked gene expression (Figure 4B). Next, we asked if the extent of X-to-A dosage compensation differs between m6A methylated vs. non-methylated transcripts in XENs, TSCs, and EpiSCs. To explore this, we profiled the active- $X^{\text{Mol}}:A^{\text{Mol}}$ ratio by segregating the X-linked transcripts into m6A methylated vs. non-methylated groups based on MeRIP-seq analysis. We find that active- $X^{\text{Mol}}:A^{\text{Mol}}$ ratio for X-linked transcripts without m6A methylation was significantly higher compared to the m6A-modified X-linked transcripts, suggesting that the extent of X-to-A dosage compensation differs between m6A methylated vs. non-methylated X-linked transcripts (Figure 4C). To test further, we profiled the allelic expression of autosomal transcripts and m6A methylated/non-methylated X-linked transcripts (Figure 4D). We find that active-X expression of both m6A and non-m6A X-linked transcripts are higher compared to the allelic autosomal expression, indicating both classes of RNAs are upregulated (Figure 4D). Interestingly, the expression of non-m6A X-linked transcripts tend to be higher compared to the m6A-modified X-linked transcripts, corroborating the fact that the extent of upregulation differs between m6A vs. non-m6A X-linked transcripts (Figure 4D). Considering MeRIP-seq analysis lacks quantitative resolution and may have false-positives or -negatives, we extended our analysis using available GLORI datasets, which provides quantitative estimation of m6A enrichment at single-base pair resolution (Liu et al., 2023). Using GLORI data, we found that transcripts with lower enrichment of m6A always tend to have higher X:A ratio compared to the transcripts with higher m6A enrichment in ESCs, MEFs, and HEK293T and HeLa cells (Figure 4E). Taken together, we conclude that the degree of X-to-A dosage compensation

negatively correlates with the m6A methylation level of X-linked transcripts.

Depletion of m6A RNA methylation has minor effects in maintaining X-to-autosome dosage compensation

It is believed that while less enrichment of m6A on X-linked transcripts increases their stability, higher enrichment on autosomal transcripts reduces the stability of autosomal RNAs and thereby contributes to the maintenance of the X-to-A dosage balance. Therefore, we investigated whether m6A RNA methylation contributes to the maintenance of the X-to-A dosage compensation in XENs, TSCs, and EpiSCs. To explore this, we compared the allelic active- $X^{\text{Mol}}:A^{\text{Mol}}$ ratio between DMSO and inhibitor-treated (m6A-depleted) XENs, TSCs, and EpiSCs (Figure 5A). Interestingly, we found that depletion of m6A RNA methylation in XENs, TSCs, and EpiSCs resulted in a slight downward shift in the active- $X^{\text{Mol}}:A^{\text{Mol}}$ ratio compared to DMSO control, however, the difference was not statistically significant in TSCs and XENs (Figure 5B). Moreover, estimation of allelic X:A ratio using the same set of genes between DMSO and inhibitor-treated groups for individual cell types also showed a similar trend (Figure S1C). To test further, we compared the overall expression of autosomal and X-linked genes between DMSO vs. m6A-depleted XENs, TSCs, and EpiSCs at an allelic resolution. We observed that expression of X-linked genes from the active-X was significantly higher compared to the autosomal allelic expression in inhibitor-treated cells (Figure 5C). Surprisingly, we did not observe any significant changes in the expression of X-linked genes on the active-X chromosome between DMSO- vs. inhibitor-treated cells (Figure 5C). However, we found significant changes in autosomal allelic expression between DMSO and inhibitor-treated cells in XEN and TSC (Figure 5C). Comparison of the cumulative distribution of log2-fold change of expression upon m6A depletion showed that X-linked transcripts are less affected compared to autosomal transcripts (Figures S2A and S2B). To further validate, we performed allelic as

Figure 4. Profiling of X-to-A dosage compensation pattern of m6A and non-m6A methylated X-linked transcripts in TSCs, XENs, and EpiSCs

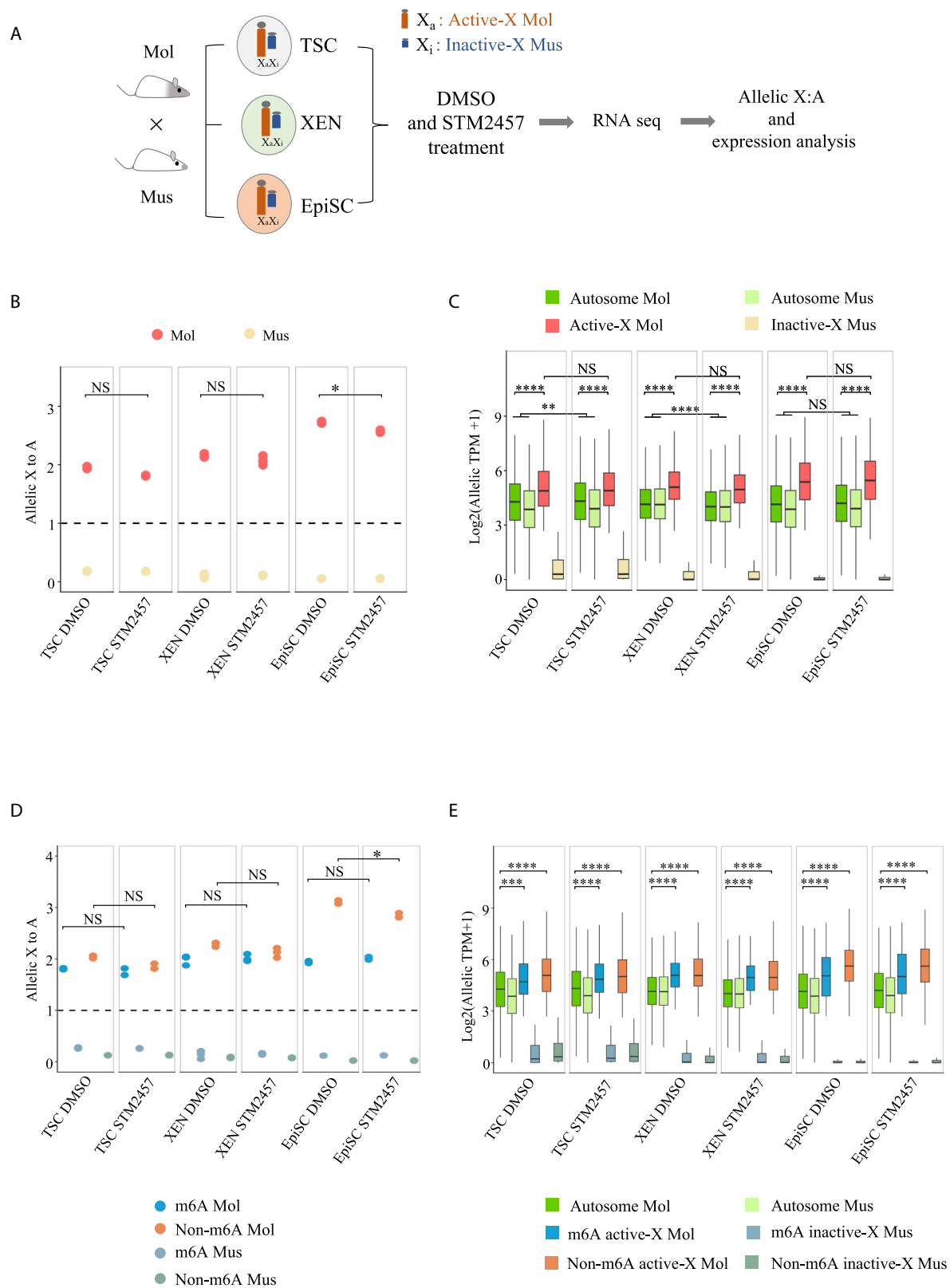
(A) Plots showing allelic X:A ratio in DMSO-treated TSCs, XENs, and EpiSCs.

(B) Boxplot showing the allelic expression of autosomal and X-linked genes in DMSO-treated TSCs, XENs, and EpiSCs; Mann-Whitney U test; **** p value <0.0001.

(C) Schematic showing analysis workflow for profiling X:A ratio for m6A and non-m6A methylated X-linked transcripts (left); allelic X:A ratio of m6A and non-m6A methylated X-linked transcripts in TSCs, XENs, and EpiSCs (right); two sample t test; * p <0.05, ** p <0.01.

(D) Boxplot showing allelic expression of autosomal and m6A-enriched or non-m6A-enriched X-linked genes in DMSO-treated TSCs, XENs, and EpiSCs; Mann-Whitney U test; ** p <0.01, *** p <0.001, **** p <0.0001; NS, nonsignificant. TSCs (n = 2), XENs (n = 3), and EpiSCs (n = 2).

(E) Plot representing comparative X:A ratio of X-linked transcripts with high vs. low level of m6A methylation as quantified through the GLORI method in different mouse cell lines (left: ESCs and MEFs) and human cell lines (right: HEK293T and HeLa).



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well as non-allelic X:A ratio analysis with 95% confidence intervals using 10,000 bootstrap replications with random sampling and replacement method (Cox et al., 1994; Rücklé et al., 2023; Sangrithi et al., 2017) (Figure S3). In consistence, we did not found any significant differences in X:A ratio upon depletion of m6A in XENs, TSCs, and EpiSCs (Figures S3A and S3B). Next, we compared active- $X^{Mol}:A^{Mol}$ ratio for m6A methylated and non-methylated X-linked transcripts in the DMSO- vs. inhibitor-treated cells. Surprisingly, we did not observe any significant differences in active- $X^{Mol}:A^{Mol}$ ratio between DMSO- vs. inhibitor-treated cells for m6A-modified X-linked transcript (Figure 5D). However, we noticed a downward shift for active- $X^{Mol}:A^{Mol}$ ratio for m6A non-methylated X-linked transcripts in inhibitor-treated cells (Figure 5D). However, the difference of active- $X^{Mol}:A^{Mol}$ ratio for m6A non-methylated X-linked transcripts between DMSO- vs. inhibitor-treated cells was not significant in TSCs and XENs (Figure 5D). But, in case of EpiSCs, we find that the difference is statistically significant (Figure 5D). Additionally, we observed that despite m6A depletion, both m6A methylated and non-methylated X-linked transcripts maintained higher allelic expression from the active-X compared to the autosomes, suggesting no effect on X-to-A dosage compensation (Figure 5E). Taken together, we conclude that depletion of m6A RNA methylation has a minor impact on X-to-A dosage equilibrium in XENs, TSCs, and EpiSCs.

Next, we extended our analysis in other cell types using available RNA-seq datasets (Geula et al., 2015; Wang et al., 2018a; 2019; 2023a; Wu et al., 2018; Xu et al., 2021; Zhao et al., 2014). We observed that m6A depletion through STM2457 treatment in ESCs and C643, HEK293T, and RPE1 cells leads to significant reduction of X:A ratio, as reported previously (Rücklé et al., 2023) (Figure 6A). However, we observed no significant alterations in X:A ratio in m6A-depleted *Mettl3* knockout (KO) or knockdown (KD) cells (ESCs, maDCs, MSCs, 3T3-L1, brain) except ESC-E14Tg-2a and liver (Figure 6B). Separately, m6A depletion in *Mettl14* KO cells resulted in significant reduction in ESCs but not in NPCs (Duda et al., 2021; Y. Wang et al., 2018b) (Figure 6C). On the other hand, we

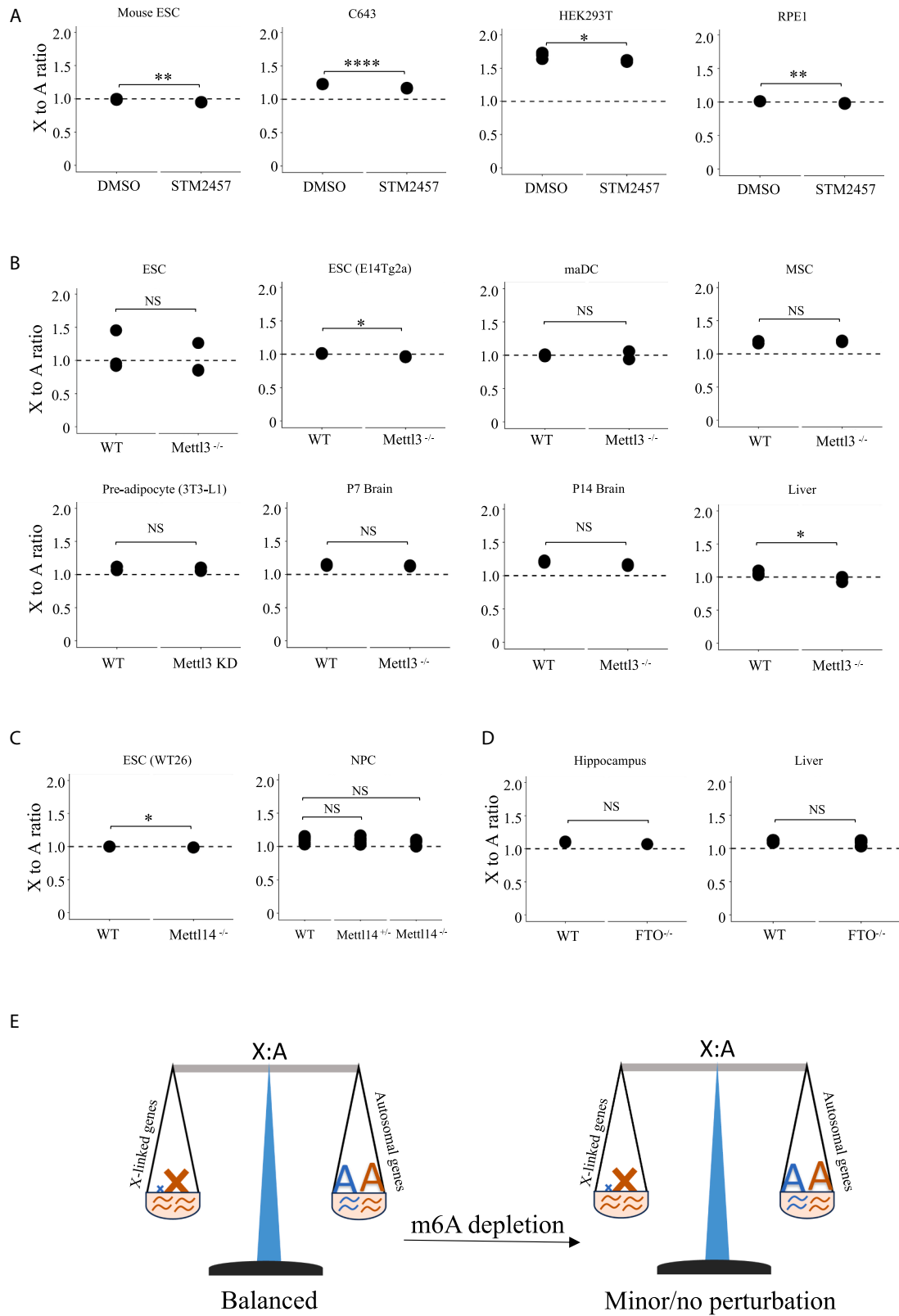
observed that ablation of m6A-eraser enzyme FTO does not alter X:A ratio in liver and hippocampus (Figure 6D) (Li et al., 2017b; Peng et al., 2019). Cumulatively, our analysis suggests that ablation of m6A-writer/eraser enzymes has minor effects on maintaining X-to-A dosage in most cases. Together, we conclude that the m6A RNA modification partly contributes to the maintenance of X-to-A dosage compensation in a cell-type-specific manner (Figure 6E).

DISCUSSION

In this study, we have dissected the role of m6A RNA methylation in dosage compensation in early embryonic lineages and somatic cells. First, we explored the role of m6A RNA methylation in the maintenance of X-linked gene silencing on the inactive-X in embryonic lineages with iXCI (XENs and TSCs) and rXCI (EpiSCs). Surprisingly, we observed that *METTL3* inhibition-mediated depletion of m6A RNA methylation in these cells does not perturb the inactive-X gene silencing, suggesting that m6A RNA methylation is dispensable for the maintenance of X-linked gene silencing (Figure 2). Our results are consistent with previous studies, which showed that *Mettl3* KO or even segmental deletion of certain m6A-containing sites on *Xist* lncRNA in mouse embryonic stem cells (mESCs) barely abrogates the gene-silencing efficiency during the initiation of XCI (Coker et al., 2020; Nesterova et al., 2019). Notably, a recent study has shown that *METTL3* depletion leads to acceleration of X-linked gene silencing, rather than perturbing it (Wei et al., 2025). On the other hand, contrasting results have been observed in mESCs, showing that *Xist* RNA methylation promotes X-linked gene silencing by facilitating the binding of several protein complexes involved in gene repression (Patil et al., 2016). Additionally, work from another group has demonstrated that m6A sites on *Xist* RNA play a pivotal role in silencing of X-linked genes on the inactive-X in HEK293T cells (Chang et al., 2022). However, a major limitation in both of their studies was that they experimented on only a few X-linked genes. Taken together, our study

Figure 5. Effect of depletion of RNA m6A levels in the maintenance of X-to-A dosage in TSCs, XENs, and EpiSCs

- (A) Schematic outline showing allelic X:A ratio analysis in the DMSO- and STM2457-treated TSCs, XENs, and EpiSCs.
 (B) Plots showing allelic X:A ratio in the DMSO- vs. STM2457-treated TSCs, XENs, and EpiSCs; two-sample *t* test; **p* < 0.05; NS, nonsignificant.
 (C) Boxplot showing allelic expression of autosomal and X-linked genes in the DMSO- and STM2457-treated TSCs, XENs, and EpiSCs; Mann-Whitney U test; ***p* < 0.01, *****p* < 0.0001; NS, nonsignificant.
 (D) Plots showing the allelic X:A ratio of m6A- and non-m6A-enriched X-linked transcripts in DMSO- vs. STM2457-treated TSCs, XENs, and EpiSCs. Two-sample *t* test; **p* < 0.05; NS, nonsignificant.
 (E) Boxplot showing allelic expression level of autosomal and X-linked genes (m6A and non-m6A enriched) in DMSO- and STM2457-treated TSCs, XENs, and EpiSCs; Mann-Whitney U test; ****p* < 0.001, *****p* < 0.0001. TSCs (*n* = 2), XENs (*n* = 3), and EpiSCs (*n* = 2).



(legend on next page)



demonstrates that m6A RNA methylation is dispensable for the maintenance of X-linked gene silencing on the inactive-X in XENs, EpiSCs, and TSCs. However, one of the limitations of our study is that we performed inhibitor treatment for 12 h to deplete the m6A level; therefore, in the future, prolonged depletion of m6A may provide better clarity into the requirement of m6A RNA methylation in the maintenance of X-linked gene silencing on the inactive-X chromosome.

On the other hand, we demonstrate that during early embryogenesis, starting from the 2-cell stage to the blastocyst stage, X-linked transcripts are less m6A methylated compared to the autosomal transcripts (Figure 3A). The trend of reduced m6A levels on X chromosome transcripts was consistent in different embryonic lineages such as XENs, EpiSCs, TSCs, and ESCs (Figures 3B, 3C, and 3E). Most of the other cell types we analyzed such as MEFs, NPCs, MSCs, dendritic cells, and liver tissues also exhibited similar trends (Figures 3D and 3E). Taken together, we conclude that reduced enrichment of m6A modification on X-linked transcripts compared to the autosomal transcripts is consistent across most of the embryonic or somatic cell types that we have analyzed. Previously, it was demonstrated that RNA-mediated regulation of differential transcript stability between X chromosomes vs. autosomes contributes to modulating the transcriptional dosage between monoallelic X-linked genes and biallelic autosomal genes (Deng et al., 2013). Indeed, a recent study has shown that preferential destabilization of autosomal transcripts over X-linked transcripts is the result of higher m6A enrichment on autosomal transcripts. Moreover, they showed that the X-linked transcripts have evolved to bear lesser m6A motifs and thereby, have lesser m6A enrichment intrinsically compared to autosomal RNAs (Rücklé et al., 2023). Overall, our data support their findings and demonstrate the universality of the observation in different cellular contexts. On the other hand, we found that the extent of X-to-A dosage compensation negatively correlates with the m6A methylation level of the X-linked transcripts (Figure 4). While MeRIP-seq analysis in XENs, EpiSCs, and TSCs suggests this trend, it may include false positives/negatives to some extent due to potential batch effects and it lacks quantitative resolution. To validate our findings, we used GLORI, which provides a quantitative

estimation of methylation level at single-base pair resolution. Consistent with MeRIP-seq, GLORI confirmed that transcripts with a lower m6A methylation tend to show higher X:A ratios (Figure 4E).

Next, we investigated whether m6A is required for the maintenance of X-to-A dosage compensation in XENs, TSCs, and EpiSCs. Our analysis revealed that depletion of global m6A in XENs, TSCs, and EpiSCs leads to a slight decrease in active-X^{Mol}:A^{Mol} ratio, suggesting that m6A RNA methylation contributes partly to maintaining the X-to-A dosage in XENs, TSCs, and EpiSCs (Figure 5B). However, we find that the magnitude of the reduction in X:A ratio upon m6A depletion in XENs, TSCs, and EpiSCs varied. While m6A-depleted XENs and TSCs showed a slight decrease in X:A ratio, EpiSCs showed a significant reduction, suggesting m6A loss impacts the X-to-A dosage compensation in a lineage-specific manner (Figure 5B). On the other hand, we found that there were no significant changes in X-linked gene expression between inhibitor- and DMSO-treated cells; however, there were alterations in autosomal gene expression between the inhibitor- and DMSO-treated cells, suggesting that the loss of m6A destabilizes the X-to-A dosage balance mainly by affecting the autosomal gene expression (Figures 5C, S2A, and S2B). Separately, although we observed a significant global depletion of m6A upon 12-h inhibitor treatment, we believe that a more stringent strategy of prolonged m6A or stable m6A depletion may provide better clarity on the contribution of m6A RNA methylation to the maintenance of X-to-A dosage compensation in XENs, TSCs, and EpiSCs. Next, we extended our analysis to different embryonic and somatic cell types using different available datasets. Consistent with the previous report, we find that m6A inhibition in ESCs and C643, HEK293T1, and RPE1 cells leads to a significant reduction of X:A ratio; however, the magnitude of the effect was small (Figure 6A) (Rücklé et al., 2023). On the other hand, despite the depletion of m6A RNA methylation in Mettl3 and Mettl14 deletion or KD cells, there was no significant alteration in the X:A ratio in most cases (Figures 6B and 6C). However, we would like to mention that in many cases, we observed a slight downward shift in the X:A ratio in m6A-depleted cells. Taken together, we demonstrate that m6A RNA methylation is required but has a minor role in the maintenance of X-to-A

Figure 6. Effect of depletion of RNA m6A levels on X-to-A dosage balance in various cell types and tissues

- (A) Plots showing the X:A ratio in the DMSO- vs. STM2457-treated mouse ESCs and human cell lines (C643, HEK293T, and RPE1). Two-sample *t* test; **p* < 0.05, ***p* < 0.01, *****p* < 0.0001.
- (B) Comparison of X:A ratio between wild-type (WT) and Mettl3 KO or KD mouse cells and tissues; two-sample *t* test; **p* < 0.05; NS, nonsignificant.
- (C) Comparison of X:A ratio between WT and Mettl14 KO mouse cells; two-sample *t* test; **p* < 0.05; NS, nonsignificant.
- (D) Comparison of X:A ratio between WT and FTO KO mouse hippocampus and liver; two-sample *t* test; NS, nonsignificant.
- (E) Model diagram depicting the effect of global depletion of m6A level on X-to-A dosage compensation.



dosage compensation. On the other hand, gain of m6A enrichment in *Fto* (m6A-eraser) KO hippocampus or liver cells did not alter the X:A ratio (Figure 6D). It is worth mentioning that in some of the datasets that we have analyzed, we do not have clarity about the efficiency of loss or gain of m6A enrichment such as *Mettl3* KD in 3T3-L1 and MEF cells (Zhao et al., 2014), *Mettl14* KO in WT26 ES cell lines (Duda et al., 2021), and *Fto* KO in mouse liver (Peng et al., 2019). In summary, our study provides significant insight into the role of m6A RNA methylation in regulating dosage compensation. We demonstrate that m6A RNA modification is dispensable for the maintenance of XCI and contributes partially to balancing the X-to-A dosage in a cell-type-specific manner.

METHODS

Allele-specific RNA-seq analysis

Allelic expression profile of X-linked and autosomal genes was obtained through analysis of the RNA-seq data using an allele-specific expression analysis pipeline based on strain-specific SNPs as described previously (Ayyamperumal et al., 2024; Majumdar et al., 2025; Naik et al., 2021, 2022, 2024). We obtained strain-specific SNPs (*Mus musculus musculus* [Mus] and *Mus musculus molossinus* [Mol]) from the Mouse genome project (<https://www.sanger.ac.uk/science/data/mouse-genomes-project>). Next, we created strain-specific *in silico* reference genomes using the variant calling file tool (Danecek et al., 2011). Basically, we generated strain-specific *in silico* reference genomes by incorporating strain-specific SNPs into the GRCm39 reference genome. Next, transcriptomic reads were mapped to the strain-specific *in silico* reference genomes using STAR (–outFilterMultiMapNmax 1). Following the mapping of the reads, SNP-wise counts were generated. After generating SNP-wise read counts, we applied the following filters to remove any false-positives in our allelic counts. First, SNPs with minimum read counts of 10 per SNP site were only selected for further analysis. Next, to obtain the allelic count for individual genes, we considered only those genes that had at least two SNPs that qualified the above SNP threshold. Additionally, the allelic read counts for individual genes were deduced by taking an average of SNP-wise reads. Finally, the allelic ratio was calculated using the following formula: (Mol-allele or Mus-allele)/(Mol-allele+Mus-allele).

X-to-A ratio analysis

X-to-A expression ratio was estimated following the pipeline as described previously (Naik et al., 2022, 2024; Pacini et al., 2021). First, we defined gene sets for both X chromosome and autosomes by excluding lowly expressed genes (<5 TPM) from our analysis. Similarly, highly expressed genes

were removed from autosomes and X chromosomes by imposing an upper TPM threshold (95th percentile). Taking the filtered gene set, X:A ratio was calculated using the bootstrapping approach. Basically, X-linked gene expression (TPM) was divided with the same number of autosomal genes selected randomly and was repeated 1,000 times to mitigate the differences between a large number of autosomal genes compared to the small number of X chromosome genes. Finally, X-to-A ratio was obtained from the median value of 1,000 repeats. For further validation, additional X:A analysis was performed for XENs, EpiSCs, and TSCs (DMSO vs. STM2457) using a bootstrapping approach based on random sampling from a distribution with replacement as described previously (Cox et al., 1994; Rücklé et al., 2023; Sangrithi et al., 2017). In brief, the X:A ratio with 95% confidence intervals was estimated through pairwiseCI from R package using function Median.ratio with 10,000 bootstrap events.

Statistical analysis and plots

Plots were generated using R packages ggplot2, Complex Heatmap, and python package pyGenomeTrackstracks (Lopez-Delisle et al., 2021). For statistical analysis, we used two-sample *t* tests and Mann-Whitney U test.

RESOURCE AVAILABILITY

Lead contact

Further information and requests should be directed to Dr. Srimonta Gayen: srimonta@iisc.ac.in.

Materials availability

This study did not generate any new unique reagents.

Data and code availability

The accession number for the RNA-seq and MeRIP-seq datasets reported in this paper are GEO: GSE295196; and GEO: GSE295197. Details about the previously available RNA-seq, MeRIP-seq, and GLORI datasets are provided in Table S1.

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AUTHOR CONTRIBUTIONS

Conceptualization, S.G., H.C.N., R.B., and A.S.; supervision and funding, S.G.; funding: experiments, data analysis, methodology, and resources, H.C.N., R.B., M.N., and K.S.; writing – first draft, A.S., R.B., and H.C.N.; editing the manuscript, S.G., A.S., H.C.N., and R.B.; the final manuscript was approved by all the authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

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